Time-Aware Inertial Normalization for Irregularly-Sampled Tabular Streams

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Abstract

Standard normalization layers in neural networks are time-blind: the Exponential Moving Average (EMA) that maintains running statistics applies a fixed forgetting weight α regardless of the elapsed time between observations.

We introduce **Time-Aware Inertial Normalization (TAIN)**, which replaces the fixed EMA coefficient with $\alpha^{\Delta t}$, where Δt is the real time gap between consecutive observations. The formula $\alpha^{\Delta t} = e^{-\lambda \Delta t}$ is well established for exponential smoothing of irregular time series [Wright, 1986, Zumbach & Müller, 2001]; our contribution is applying it to the *normalization layer's running statistics*, a component that all prior irregular time series methods (GRU-D, Neural ODEs, mTAN) leave untouched. We show that $\alpha^{\Delta t}$ is the natural discretization of the Ornstein-Uhlenbeck process, grounding TAIN in continuous-time stochastic process theory. We validate TAIN empirically as a running-statistics tracker across five real-world domains; end-to-end integration as a drop-in BatchNorm replacement in tabular neural network training is left to future work.

To evaluate TAIN's impact on action stability, we analyze jitter using a decomposition into *signal jitter* (desirable tracking of genuine signal movement) and *unnecessary jitter* (undesirable oscillation exceeding signal change), combined with flip rate and directional agreement metrics drawn from signal processing and econometrics. No prior work on policy churn [Schaul et al., 2022] or action oscillation [Chen et al., 2021] provides this decomposition.

Empirical validation on five real-world datasets (Retail, Sensor, Finance, ICU-Temp, ICU-Urine; 5,409 entities, 659,325 observations) demonstrates that TAIN achieves statistically significant RMSE improvements over standard EMA ($p < 0.001$ in four of five domains). An L^1 jitter decomposition shows that the bulk of TAIN's jitter increase over Standard EMA is signal-tracking rather than spurious (80–98% signal across the five domains), and that TAIN's unnecessary-share inflation over EMA is the smallest among RMSE-improving methods. Stratified post-gap analysis shows that the largest gap stratum yields the largest improvement in four of five domains, with the sensor domain exhibiting a clean monotonic increase from +7.6% to +67.6% across strata — the strongest direct confirmation of the Ornstein-Uhlenbeck prediction. Within-domain non-monotonicities (e.g., the Rossmann 2–3 day reversal) and entity-level heterogeneity (Section 5.2.7) reveal limits of a single global α.

Code is available at https://github.com/segmora/TAIN.

1. Introduction

Machine learning on tabular data has long been dominated by two families of models: gradient-boosted decision trees (GBDT) and multilayer perceptrons (MLP). Both share a fundamental assumption: observations arrive independently, at uniform intervals, with the sole objective of minimizing a pointwise prediction loss. This assumption fails systematically in operational AI settings.

Consider a retail chain deploying a store management intelligence system. Daily sales figures arrive on weekdays, but weekend data is aggregated; public holiday gaps of 7–14 days interrupt otherwise regular streams; sensor outages introduce sporadic missingness. The temporal structure of the data (the fact that a 30-day gap carries different information than a 1-day gap) is invisible to any model that treats each record as an i.i.d. sample.

The consequence goes beyond suboptimal forecasting. The downstream objective is not prediction but *action*: a store manager receives a daily directive (reorder this SKU, redeploy staff, trigger a promotion). When the normalization layer is time-blind, two observations separated by a holiday gap are treated identically to two observations separated by a single day. The model's internal statistics carry stale momentum; the generated actions oscillate. We call this **action jitter**, and it is the primary cause of trust erosion in operational AI deployments.

This paper addresses two gaps in the literature. First, while methods such as GRU-D [Che et al., 2018], Neural ODEs [Chen et al., 2018], and mTAN [Shukla & Marlin, 2021] introduce time-awareness into hidden state dynamics or attention mechanisms, *none modifies the running-statistics update underlying the normalization layer*. The running statistics that BatchNorm maintains via EMA are updated with a fixed α regardless of observation spacing. Second, while policy churn [Schaul et al., 2022] and action oscillation [Chen et al., 2021] have been identified as problems, no standard methodology decomposes jitter into its desirable (signal-tracking) and undesirable (noise-driven) components.

1.1 Contributions

- **TAIN**: a time-aware running-statistics update rule for the normalization layer, in which the inertia coefficient $\alpha^{\Delta t}$ adapts to the real elapsed time between observations, derived as the natural discretization of the Ornstein-Uhlenbeck process. The formula $\alpha^{\Delta t}$ is well known in time series analysis [Wright, 1986, Zumbach & Müller, 2001]; our contribution is its application to the running-statistics update, a component untouched by prior irregular time series methods. We validate TAIN as a tracker; end-to-end integration as a drop-in BatchNorm replacement is left to future work.

- **Jitter decomposition analysis**: we evaluate TAIN using a decomposition of total estimate jitter into signal jitter (desirable) and unnecessary jitter (undesirable), complemented by flip rate and directional agreement, all drawn from signal processing and econometrics. This analytical lens enables principled evaluation of the RMSE–jitter trade-off.

- **Empirical validation** on five real-world datasets (5,409 entities, 659,325 observations) against six baselines including the Kalman Filter, Holt Exponential Smoothing, and Double EMA. TAIN is shown to achieve the best RMSE and post-gap recovery within the low-jitter operating regime.

2. Background and Related Work

2.1 Neural ODEs and Irregular Time Series

Neural ODEs [Chen et al., 2018] define the hidden state $\mathbf{h}(t)$ as a continuous-time quantity governed by a learned vector field:

$$\frac{d}{dt}\mathbf{h}(t) = f(\mathbf{h}(t), \mathbf{x}(t), \theta), \tag{1}$$

where $\mathbf{x}(t)$ is the input at time t and θ are learnable parameters. The continuous-time formulation handles irregular observation times naturally. Latent ODEs [Rubanova et al., 2019] extend this to sequences with structured missingness. Neural CDEs [Kidger et al., 2020] extend the formulation to controlled dynamics driven by the data path. Liquid Time-Constant Networks [Hasani et al., 2021] introduce input-dependent time constants into ODE-based recurrent architectures. Multi-Time Attention Networks [Shukla & Marlin, 2021] address irregular sampling through continuous-time attention. GRU-D [Che et al., 2018] handles missing values via exponential decay on the hidden state.

All of these methods introduce time-awareness into the *hidden state dynamics* or the *attention mechanism*, yet none modifies the normalization layer's running statistics. The EMA that maintains $\hat{\mu}$ and $\hat{\sigma}^2$ uses a fixed α in all of these architectures, regardless of the time gap between observations.

2.2 Batch Normalization and EMA

Batch Normalization [Ioffe & Szegedy, 2015] maintains running statistics via EMA:

$$\hat{\mu}_t = (1-\alpha)\,\mu_{\text{batch}} + \alpha\,\hat{\mu}_{t-1}, \tag{2}$$

where $\alpha \in (0, 1)$ is a *fixed* hyperparameter. In irregular-interval settings this introduces systematic bias: long gaps should produce larger resets, but standard EMA treats them identically to short gaps.

2.3 Exponential Decay for Irregular Observations

The idea of adapting the EMA coefficient to the elapsed time between observations has a long history. Wright [1986] proposed modified exponential smoothing for data published at irregular intervals, deriving the time-dependent weight from classical Holt's method. Zumbach & Müller [2001] developed a comprehensive framework of operators on inhomogeneous time series for quantitative finance, with the exponential kernel $e^{-\Delta t/\tau}$ as the fundamental building block. Cipra [2006] extended these ideas to higher-order exponential smoothing.

The formula $\alpha^{\Delta t} = e^{-\lambda \Delta t}$ is thus well established. What has *not* been proposed is applying this formula to the normalization layer's running statistics in a neural network. This is the specific gap that TAIN addresses.

2.4 Time-Aware Batch Normalization

The closest prior work is TA-BN [Choi et al., 2024], which associates separate population statistics with predefined time grids for Neural ODE integration. However, TA-BN addresses "time" in the sense of ODE integration depth (a continuous-depth parameter), not the real-world elapsed time between irregularly arriving observations. TAIN addresses a different notion of "time": the actual temporal gap between data points in an operational stream.

2.5 Action Stability and Jitter

Schaul et al. [2022] identify *policy churn* (the fraction of states whose greedy action changes after a single batch update) as a pervasive phenomenon in deep RL. Chen et al. [2021] address action oscillation in offline RL via a policy inertia controller. Mysore et al. [2021] propose temporal and spatial smoothness regularization for continuous control.

These works establish that action instability is a problem, but none provides a *decomposition* of jitter into signal-tracking (desirable) and unnecessary (undesirable) components. In operational AI, not all jitter is bad: estimate changes that correctly track genuine demand shifts are essential. The jitter decomposition analysis in Section 3.2 addresses this gap by combining established metrics into a coherent evaluation methodology.

3. Method

3.1 Time-Aware Inertial Normalization (TAIN)

3.1.1 Formulation

Let $\hat{\mu}_t$ denote the running mean at observation time t, μ_{batch} the mean of the current batch, and $\Delta t = t - t_{\text{prev}}$ the elapsed real time. TAIN replaces the fixed EMA coefficient α with $\alpha^{\Delta t}$:

$$\boxed{\hat{\mu}_t = \left(1 - \alpha^{\Delta t}\right) \mu_{\text{batch}} + \alpha^{\Delta t}\, \hat{\mu}_{t-1}} \tag{3}$$

$$\hat{\sigma}_t^2 = \left(1 - \alpha^{\Delta t}\right) \sigma_{\text{batch}}^2 + \alpha^{\Delta t}\, \hat{\sigma}_{t-1}^2 \tag{4}$$

$$\mathbf{h}_{\text{norm}} = \frac{\mathbf{h} - \hat{\mu}_t}{\sqrt{\hat{\sigma}_t^2 + \varepsilon}} \tag{5}$$

Remark 1. *The formula $\alpha^{\Delta t}$ is equivalent to the exponential kernel $e^{-\lambda \Delta t}$ used by Wright [1986] and Zumbach & Müller [2001] for irregularly-sampled time series smoothing. Our contribution is not the formula itself but its application to the running-statistics update used in normalization layers, where prior work universally uses a fixed α. We validate this application empirically as a tracker; end-to-end integration in neural network training is left to future work (Section 6.4).*

3.1.2 Properties

The $\alpha^{\Delta t}$ substitution has three limiting behaviors:

- $\Delta t \to 0$: $\alpha^{\Delta t} \to 1$ — full inertia preserved across very short gaps.
- $\Delta t = 1$: $\alpha^1 = \alpha$ — reduces to standard EMA.
- $\Delta t \to \infty$: $\alpha^{\Delta t} \to 0$ — full reset; long gaps treated as fresh starts.

A 30-day shop closure should reset internal statistics toward current conditions; a 1-hour gap within a trading day should preserve accumulated inertia.

3.1.3 Connection to Ornstein-Uhlenbeck Process

Writing $\alpha = e^{-\lambda}$ for decay rate $\lambda > 0$:

$$\hat{\mu}_t = \left(1 - e^{-\lambda \Delta t}\right) \mu_{\text{batch}} + e^{-\lambda \Delta t}\, \hat{\mu}_{t-1} \tag{6}$$

This is the exact discrete-time solution of the Ornstein-Uhlenbeck (OU) SDE [Uhlenbeck & Ornstein, 1930]:

$$d\hat{\mu} = -\lambda\,(\hat{\mu} - \mu_{\text{target}})\,dt + \sigma\,dW_t \tag{7}$$

in the deterministic limit ($\sigma = 0$), where $\mu_{\text{target}} = \mu_{\text{batch}}$. TAIN is the natural discretization of a mean-reverting continuous-time process for irregular observation schedules. The hyperparameter α has a principled interpretation: the per-unit-time decay rate of the running statistics.

While the OU discretization has been used for time series smoothing since Wright [1986], its application to normalization layer running statistics is, to our knowledge, novel. The OU connection provides theoretical grounding for why $\alpha^{\Delta t}$ is the *correct* time adaptation rather than a heuristic: it preserves the mean-reverting dynamics of the underlying continuous process regardless of observation spacing.

3.2 Jitter Decomposition Analysis

Total jitter — whether measured by RMS or by mean absolute change — conflates two distinct phenomena: estimate changes that track genuine signal movement (desirable) and estimate changes that exceed the signal change (undesirable). We decompose the latter using the L^1 (mean-absolute) norm because, unlike the RMS form, it admits an exact additive identity.

Definition 1 (L^1 Jitter Decomposition). *Let $\Delta\hat{\mu}_t = \hat{\mu}_t - \hat{\mu}_{t-1}$ be the estimate change and $\Delta x_t = x_t - x_{t-1}$ the signal change. Define the L^1* total jitter, signal jitter, *and* unnecessary jitter*:*

$$J^{L^1}_{total} = \tfrac{1}{N-1}\textstyle\sum_t |\Delta\hat{\mu}_t| \tag{8}$$

$$J^{L^1}_{signal} = \tfrac{1}{N-1}\textstyle\sum_t \min\bigl(|\Delta\hat{\mu}_t|,\ |\Delta x_t|\bigr) \tag{9}$$

$$J^{L^1}_{unnecessary} = \tfrac{1}{N-1}\textstyle\sum_t \max\bigl(0,\ |\Delta\hat{\mu}_t| - |\Delta x_t|\bigr) \tag{10}$$

These satisfy the exact identity $J^{L^1}_{total} = J^{L^1}_{signal} + J^{L^1}_{unnecessary}$, so the unnecessary share *$J^{L^1}_{unnecessary}/J^{L^1}_{total}$ is a strict fraction. The standard RMS jitter $J^{RMS}_{total} = \sqrt{(N-1)^{-1}\sum_t \Delta\hat{\mu}_t^2}$ remains the operational time-series stability metric and is reported alongside the L^1 components.*

Remark 2. *A naive RMS decomposition with $J^{RMS2}_u = (N-1)^{-1}\sum_t \max(0, |\Delta\hat{\mu}_t| - |\Delta x_t|)^2$ is not additive: $J^{RMS2}_t \neq J^{RMS2}_s + J^{RMS2}_u$ due to a non-vanishing cross-term in the pointwise sum $|\Delta\hat{\mu}_t|^2 = (\min + \max)^2 = \min^2 + 2\min\cdot\max + \max^2$. The L^1 form avoids this by aggregating linearly rather than quadratically.*

Definition 2 (Flip Rate and Directional Agreement). *The* flip rate *is the fraction of consecutive estimate changes that reverse direction: $Flip = \frac{1}{N-2}\sum_t \mathbb{1}[sign(\Delta\hat{\mu}_t) \neq sign(\Delta\hat{\mu}_{t-1})]$.* Directional agreement *is the fraction of estimate movements that match the signal direction: $DirAgr = \frac{1}{N-1}\sum_t \mathbb{1}[sign(\Delta\hat{\mu}_t) = sign(\Delta x_t)]$.*

Together, these metrics ($J^{L^1}_{\text{total}}$, $J^{L^1}_{\text{signal}}$, $J^{L^1}_{\text{unnecessary}}$, unnecessary share, flip rate, directional agreement) form a complete picture of action stability that total jitter alone cannot provide. A method may have high $J^{L^1}_{\text{total}}$ but low unnecessary share (correctly tracking a volatile signal); another may have low $J^{L^1}_{\text{total}}$ but high flip rate (oscillating around a stationary signal).

4. Theoretical Properties

4.1 TAIN Convergence

Let $\varepsilon_t = \hat{\mu}_t - \mu^*$ denote estimation error. Under the OU connection (Section 3.1.3):

$$\mathbb{E}[\varepsilon_t] = e^{-\lambda t_{\text{cum}}} \cdot \varepsilon_0 \tag{11}$$

where $t_{\text{cum}} = \sum_i \Delta t_i$ is total elapsed real time. TAIN's error decays exponentially in *real time*, not observation count; a more appropriate behavior for irregular-interval data.

Proposition 1 (Gap-Proportional Recovery)**.** *For two gaps $\Delta t_1 < \Delta t_2$, TAIN's post-gap weight satisfies $\alpha^{\Delta t_1} > \alpha^{\Delta t_2}$, producing a strictly larger reset for the longer gap. The recovery magnitude is monotonically increasing in Δt.*

This property is directly testable and forms the basis of our stratified post-gap recovery experiment (Section 5.2.6).

5. Experiments

5.1 Illustrative Synthetic Simulation

5.1.1 Setup

We construct a controlled synthetic retail stream of 100 observations over ≈130 real days: weekday gaps $\Delta t = 1$, weekend gaps $\Delta t = 2$, holiday gaps $\Delta t \in \{7, 9, 14\}$. True signal: seasonal sinusoid with linear growth and a structural regime shift at observation 65. Heteroscedastic noise: $\sigma = 200$ (baseline), 800 and 900 (turbulent windows).

5.1.2 Results

Table 1: TAIN vs. alternatives on irregular synthetic retail stream. RMSE evaluated against true underlying signal.

Method	RMSE ↓	Jitter ↓	Note
Batch Norm	285.4	99.5	Time-blind
Standard EMA	603.4	36.9	Fixed α
TAIN ($\alpha^{\Delta t}$)	**534.8**	46.9	11.4% RMSE ↓ vs EMA

TAIN's RMSE improvement is most visible around holiday gaps ($\Delta t = 9, 14, 7$), where standard EMA carries stale statistics. TAIN applies $\alpha^9 = 0.63$, $\alpha^{14} = 0.46$, $\alpha^7 = 0.70$, enabling faster recovery. The higher jitter (46.9 vs. 36.9) is expected: TAIN makes larger post-gap corrections to track the true signal. Jitter decomposition (Section 5.2.9) shows that this additional jitter is predominantly signal-tracking, not unnecessary oscillation.

5.2 Real-World Empirical Validation

5.2.1 Datasets

To complement the synthetic simulation in Section 5.1, we validate TAIN on five domain settings drawn from four publicly-available real-world data sources, all with naturally occurring irregular sampling. The five settings are Retail (Rossmann), Sensor (Beijing Multi-Site Air Quality), Finance (US Equities), and two channels from the PhysioNet 2012 ICU cohort: ICU-Temp and ICU-Urine. We treat ICU-Temp and ICU-Urine as separate settings because they capture distinct physiological processes (body temperature vs. urine output) on overlapping but non-identical patient subsets (1,787 vs. 3,555 patients); however, they share the same recording infrastructure and underlying cohort, so the two should not be regarded as fully independent corroborating domains. All code and data are publicly available for reproducibility.

Retail. The Rossmann dataset contains daily sales for 1,115 German drugstores over 2.5 years. Stores are closed on Sundays and public holidays, creating gaps of 2–7+ days. We select the 50

Table 2: Real-world datasets used for TAIN validation.

Domain	Dataset	Entities	Obs.	Unit	Gap Mechanism
Retail	Rossmann Store Sales	50	29,848	Daily	Store closures
Sensor	Beijing Air Quality	12	411,726	Hourly	Sensor outages
Finance	US Equities (5 tickers)	5	12,580	Daily	Weekends, holidays
ICU-Temp	PhysioNet 2012 (Temp)	1,787	61,161	Hourly	Clinical schedule
ICU-Urine	PhysioNet 2012 (Urine)	3,555	130,508	Hourly	Clinical schedule

stores with the most observations and filter to open days with positive sales, yielding irregular time series with a mean of 119 gaps per store.

Sensor. The Beijing Multi-Site Air Quality dataset [Zhang et al., 2017] provides hourly PM2.5 readings from *all 12* monitoring stations in the public release (Aotizhongxin, Changping, Dingling, Dongsi, Guanyuan, Gucheng, Huairou, Nongzhanguan, Shunyi, Tiantan, Wanliu, Wanshouxigong) over 4 years (March 2013–February 2017); no station subset was selected. Sensor outages and maintenance windows produce gaps ranging from 2 hours to 350+ hours, with a total of 2,837 gaps across all stations.

Finance. Daily closing prices for five large-cap US technology equities (AAPL, MSFT, GOOGL, AMZN, TSLA) from January 2015 through December 2024, sourced from Yahoo Finance. These tickers were selected as a common benchmark subset (a 5-ticker slice of the so-called "Magnificent 7") prior to running TAIN; they were not screened for favorable behavior under the method. Weekend closures produce 3-day gaps; market holidays produce 4-day gaps. Each ticker contains approximately 544 gaps. We acknowledge that $N = 5$ tickers is a small sample and report Finance results with this caveat in mind; expanding to a broader equity universe is a natural extension.

ICU-Temp. The PhysioNet 2012 Challenge dataset [Silva et al., 2012] contains ICU temperature readings from 4,000 patients. Measurements are taken at clinically-determined intervals, producing genuinely irregular sampling with 22.1% of observations following gaps > 1.5 hours. We retain the 1,787 patients with ≥ 15 temperature readings.

ICU-Urine. From the same PhysioNet 2012 dataset, urine output measurements for 3,555 patients with ≥ 15 observations. Gap rate is 18.2%, with gaps ranging from minutes to 33 hours, driven by nursing schedules and clinical protocols.

5.2.2 Methods Compared

We compare TAIN against six baselines spanning three categories, plus report TAIN's own results:

- **Time-blind smoothers:** Standard EMA (α fixed), Double EMA (DEMA: $2 \cdot \text{EMA} - \text{EMA}(\text{EMA})$), and Holt's Exponential Smoothing (level + trend).
- **Naive time-aware:** Linear-scaled EMA ($\alpha_{\text{eff}} = \min(\alpha \cdot \Delta t, 0.999)$) and Interpolation+EMA (linearly fill gaps, then apply standard EMA).
- **Classical optimal:** Kalman Filter with process noise $Q = Q_{\text{base}} \cdot \Delta t$ (time-aware by construction).
- **TAIN variant:** TAIN ($\alpha^{\Delta t}$).

All methods use $\alpha = 0.95$ as the base smoothing parameter. No method-specific hyperparameter tuning is performed.

5.2.3 Metrics

We evaluate four primary metrics:

- **Tracking RMSE:** $\sqrt{\frac{1}{N}\sum_t (x_t - \hat{\mu}_t)^2}$ — how accurately the running estimate tracks the signal.
- **Action Jitter:** $\sqrt{\frac{1}{N-1}\sum_t (\hat{\mu}_t - \hat{\mu}_{t-1})^2}$ — the magnitude of estimate changes between consecutive observations.
- **Jitter-RMSE Ratio (J/R):** Jitter divided by RMSE. Lower values indicate smoother tracking; higher values indicate volatile estimates.
- **Post-Gap MAE:** Mean absolute error in the $k = 5$ observations following a gap ($\Delta t > 1.5$), measuring recovery speed.

Additionally, we apply the jitter decomposition analysis (Section 3.2) to separate signal from unnecessary jitter.

5.2.4 Results

Table 3 reports the mean metrics across entities for each domain. Two findings are immediately apparent.

Finding 1: TAIN achieves the best RMSE and post-gap recovery within the low-jitter operating regime in four of five domains. Among methods with J/R < 0.10 (the operational regime where action stability is acceptable), TAIN achieves the lowest RMSE *and* the lowest post-gap MAE in Retail, Sensor, Finance, and ICU-Urine. In ICU-Temp, TAIN's J/R (0.118) marginally exceeds the threshold; Interp+EMA wins the strictly $J/R < 0.10$ class there on both RMSE (0.73) and post-gap MAE (0.57), while TAIN (0.75, 0.51) achieves the lowest post-gap MAE among all methods with $J/R < 0.15$. Compared to Standard EMA, TAIN reduces RMSE by 1.05% (Retail, $p < 0.001$), 0.62% (Sensor, $p < 0.001$), 17.32% (Finance, $p = 0.031$), 3.04% (ICU-Temp, $p < 0.001$), and 3.78% (ICU-Urine, $p < 0.001$). TAIN's total jitter is *higher* than Standard EMA (e.g., 120.3 vs. 98.9 in Retail) because its post-gap corrections are deliberately larger; the L^1 jitter decomposition (Section 5.2.9) shows this additional jitter is predominantly signal-tracking.

Finding 2: Higher-RMSE methods produce unacceptable jitter. The Kalman Filter achieves lower absolute RMSE in all domains but does so at 3–9× higher J/R ratio (0.25–0.52 vs. TAIN's 0.06–0.12). DEMA achieves intermediate RMSE at 1.5–2× higher J/R. Holt ES is uniformly inferior: its RMSE is the worst in every domain (e.g., 2521.4 vs. EMA's 1878.0 in Retail) *and* its J/R ratio sits in the 0.15–0.23 range, on par with Kalman and DEMA. In operational deployments where each estimate change triggers a downstream action (reorder, staff redeployment, promotion trigger), this level of jitter is prohibitive.

This result positions TAIN not as a universal RMSE minimizer but as an *optimal time-aware normalization within the action-stable operating regime*, which is where operational AI systems must operate.

5.2.5 Statistical Significance

Table 4 reports Wilcoxon signed-rank tests and bootstrap 95% confidence intervals for TAIN's RMSE improvement over standard EMA.

The improvement is statistically significant ($p < 0.05$) in all five domains, with 3,684 of 5,409 entities (68%) showing positive improvement. The win rate is not uniform across domains:

Retail (80%), Sensor (100%), and Finance (100%) show high win rates, while ICU-Urine reaches 71.4% (2,539/3,555) and ICU-Temp only 60.9% (1,088/1,787) — barely above coin flip at the per-patient level despite a highly significant aggregate p-value. This thin ICU-Temp margin is consistent with the entity-level heterogeneity discussed in Section 5.2.7: TAIN's benefit is concentrated on the acutely-unstable subset of patients rather than distributed uniformly across the cohort.

5.2.6 Stratified Post-Gap Recovery

The theoretical framework predicts that TAIN's advantage should grow with gap size, since $\alpha^{\Delta t}$ produces proportionally larger resets for larger Δt (Proposition 4.1). We test this by stratifying gaps and measuring post-gap MAE improvement at each stratum across all five domains.

The Sensor domain provides the cleanest confirmation of the theoretical prediction. Improvement increases monotonically across all five strata (7.6% → 67.6%), spanning more than 8× in magnitude. The relationship is near-linear in the log gap, consistent with the Ornstein-Uhlenbeck discretization $\alpha^{\Delta t} = e^{-\lambda \Delta t}$ established in Section 3.1.3 — though we caution that this consistency does not by itself imply the underlying PM2.5 signal is an OU process; the discretization is the correct gap-aware update under OU dynamics, and the empirical fit indicates only that the gap-improvement relationship in Sensor is compatible with this prediction.

ICU-Temp shows a clean monotonic pattern in the well-sampled strata (+23.0% → +34.4% → +34.0% from 1.5–3h to 6–12h, with 13,967 of 13,984 gaps in these three strata). The 12–24h ($n = 16$) and 24+h ($n = 1$) strata point in the same direction but have insufficient sample to confirm.

ICU-Urine, Retail, and Finance show the predicted gap-proportional effect at the extremes but with within-domain non-monotonicities at intermediate strata. Retail exhibits a −6.4% reversal at the 2–3 day stratum, attributable to Rossmann's weekly seasonality (Sunday closure produces a systematic Monday demand shift that TAIN's $\Delta t = 2$ reset overshoots). Finance shows a U-shape (29.0%, 18.2%, 23.7%), driven by the uniform 3-day weekend gap dominating $n = 2{,}260$ and giving little gap-size variation to exploit; the 4+ day holiday stratum recovers. ICU-Urine improves from +13.4% at the shortest stratum to +43.9% at the longest, but with noise in the middle (particularly the small-n 12–24h stratum at +10.5%).

Aggregate conclusion. The largest gap stratum yields the largest improvement in four of five domains; only Finance, with limited gap-size variability, deviates from this pattern in the well-populated strata. The prediction $\alpha^{\Delta t} \to 0$ as Δt grows therefore receives strong empirical support, but the relationship is monotonic in the strict sense only in the Sensor domain. Within-domain non-monotonicities — the Retail 2–3 day reversal, the Finance weekend dominance, and the ICU-Urine intermediate noise — expose the limits of a single global α and motivate the learned $\alpha(t)$ direction discussed in Future Work.

Caveat on the Retail stratified support. In Retail, the "largest-stratum-is-best" pattern is carried entirely by the 7+ day stratum ($n = 5$ gaps, italicized in Table 5). The well-populated Retail strata are 1–2 days (+2.8%, $n = 3{,}575$) and 2–3 days (−6.4%, $n = 186$): the bulk of Retail evidence is a modest positive effect at short gaps with a small reversal at intermediate gaps. Retail therefore contributes to the four-of-five aggregate count only via the small-n extreme stratum, and should not be read as independently confirming the gap-proportional prediction.

5.2.7 Entity-Level Gap-Magnitude Correlation

The stratified analysis (Section 5.2.6) aggregates gaps *within* each domain regardless of which entity they came from. A complementary question is whether entities with sparser overall measurement schedules (larger mean Δt) benefit more from TAIN. We test this *across entities* via the Spearman rank correlation between each entity's mean gap size and its RMSE improvement.

Three of the four testable domains show the expected positive sign, with two reaching significance (Retail, ICU-Urine). The Sensor non-significance is expected at $N = 12$: even a moderate true effect would require larger samples to detect at the entity level, and the within-entity stratified pattern is unambiguously positive.

ICU-Temp shows the opposite sign ($\rho = -0.139$, $p < 10^{-15}$): patients with denser measurement schedules benefit more, not less. This contradicts the naive reading of Proposition 4.1 but is not a contradiction of the within-entity stratified result (which is monotonically positive for ICU-Temp). The mechanism is clinical: patients sampled more frequently are typically the more critically unstable cases, with rapidly varying temperatures that TAIN's faster post-gap correction tracks well; patients sampled sparsely tend to be clinically stable, where standard EMA's inertia is already appropriate. The entity-level correlation is therefore confounded by patient acuity, and reflects a real heterogeneity rather than a failure of the OU discretization.

Auxiliary verification: the confound is direct. To test this mechanism, we correlated per-patient mean Temp gap with the SAPS-I and SOFA acuity scores from PhysioNet 2012's outcomes file. Both correlations are strongly negative: $\rho = -0.281$ ($p < 10^{-32}$) for SAPS-I and $\rho = -0.375$ ($p < 10^{-58}$) for SOFA, across $N = 1{,}744$ patients with valid scores. Sicker patients are sampled more densely by a margin twice as large as the $|\rho| = 0.139$ gap-improvement correlation itself. In-hospital mortality shows only a weak association with mean gap ($r = +0.113$, $p < 10^{-5}$, $N = 1{,}787$), consistent with mortality being a whole-stay outcome whereas sampling density reflects the recorded first 48 hours. The strong SAPS-I and SOFA correlations confirm that the negative ICU-Temp Spearman reflects acutely-unstable patients (with rapidly varying temperatures and dense sampling) benefiting more from TAIN's post-gap correction, rather than a failure of the OU discretization. We return to this in Limitations.

5.2.8 Sensitivity to α

We evaluate TAIN across $\alpha \in \{0.80, 0.85, 0.90, 0.95, 0.99\}$. The improvement over Standard EMA is positive for all tested values:

This monotonic relationship in Retail and Sensor is expected: lower α means faster forgetting, so the difference between α and $\alpha^{\Delta t}$ is proportionally larger. The Finance domain shows stability across α values, reflecting the uniform gap structure (weekends are consistently 3 days).

5.2.9 Jitter Decomposition

We apply the L^1 jitter decomposition (Section 3.2) to compare action stability profiles across methods. The RMS column is the operational total-jitter metric also reported in Table 3; the L^1 columns satisfy $J^{L^1}_{\text{total}} = J^{L^1}_{\text{signal}} + J^{L^1}_{\text{unnec}}$ exactly.

Finding 3: TAIN's total jitter increase over Standard EMA is overwhelmingly signal-tracking. The L^1 decomposition makes this quantitative. In Retail, TAIN's total jitter exceeds EMA's by $\Delta J^{L^1}_{\text{total}} = 84.82 - 75.88 = 8.94$, of which 8.73 is additional signal jitter and only 0.21 is additional unnecessary jitter — a 98%/2% split. The corresponding signal/unnecessary split for TAIN's increment over EMA is 97%/3% in Sensor, 87%/13% in Finance, 95%/5% in ICU-Temp, and 81%/19% in ICU-Urine. The bulk of TAIN's added jitter therefore reflects faithful post-gap signal tracking rather than noise.

Finding 4: Among RMSE-improving alternatives, TAIN has the lowest unnecessary-share inflation over Standard EMA. The unnecessary share is near-identical between EMA and TAIN ($\Delta \leq 0.2$ pp in Retail, Sensor, ICU-Urine; -1.9 pp in ICU-Temp), with the exception of Finance (+3.6 pp), where the higher share is the price of the 17.3% RMSE gain on near-uniform weekend gaps. The competing low-jitter method Interp+EMA matches TAIN's share closely but at lower RMSE improvement. In contrast, methods that achieve lower RMSE (Kalman Filter, DEMA) raise the unnecessary share by 3–6 pp over EMA in every domain (e.g., Retail Kalman: 1.7% → 4.7%; Retail DEMA: 1.7% → 3.1%; Finance Kalman: 7.9% → 12.7%). Holt ES degrades the share dramatically across domains (e.g., 1.7% → 12.7% in Retail; 14.8% → 46.9% in ICU-Urine).

Finding 5: Kalman's higher directional agreement is paired with substantially higher flip rate. In Retail, Kalman attains 72.2% directional agreement (vs. TAIN's 66.9%) at the cost of a 38.8% flip rate (vs. TAIN's 33.5%): it responds to every fluctuation, including noise-driven reversals that TAIN correctly ignores. The same pattern holds in Sensor, Finance, and the ICU domains.

5.2.10 Figures

6. Discussion

6.1 Scope of Claims

This paper makes two types of claims. **Mathematical claims** (the OU connection, TAIN convergence, and gap-proportional recovery) are rigorous and hold under the stated assumptions. **Real-world empirical claims** (TAIN's RMSE improvement, post-gap recovery, and jitter decomposition) are validated on five public datasets (5,409 entities, 659,325 observations) with Wilcoxon significance tests and bootstrap confidence intervals.

Scope of empirical validation. TAIN is evaluated as a *running-statistics tracker*: the running mean and variance evolve under the TAIN update rule and we measure tracking RMSE, jitter, and post-gap recovery against the observed signal. We do *not* integrate TAIN as a drop-in BatchNorm replacement in end-to-end tabular neural network training (TabNet, FT-Transformer, NODE) and we do not measure downstream task gains (classification AUC, regression R^2). Such integration is the natural next step and is discussed in Future Work (Section 6.4).

The jitter decomposition (Section 5.2.9) reveals that the choice between TAIN and higher-accuracy methods (e.g., Kalman Filter) is not a question of which is "better" but which *operating regime* the deployment requires: **high-jitter-tolerant** deployments (automated trading) favor Kalman; **low-jitter-required** deployments (retail store management, where action stability drives operational trust) favor TAIN.

6.2 Operational Context: Digital Twin Architectures

In operational digital twin systems, a downstream decision layer (an LLM, a rule engine, or a human operator) consumes normalized statistics to generate store-level actions on a daily or weekly cadence. The jitter profile of the normalization layer directly determines action coherence at this boundary. A digital twin whose normalization ignores observation gaps propagates stale-momentum artifacts into the decision layer; the resulting actions oscillate and erode operator trust. TAIN's gap-proportional reset provides the statistical hygiene that downstream decision layers require: after a 3-day holiday closure, running statistics are pulled toward current conditions rather than echoing pre-closure patterns.

This matters most when the decision layer is an LLM. Language models amplify input volatility: if the normalized signal oscillates between "sales rising" and "sales falling" on consecutive days, the LLM generates contradictory directives (e.g., "launch a promotion" followed by "cancel the promotion"). By suppressing unnecessary jitter at the normalization stage, TAIN reduces the risk of such downstream hallucination and keeps the action stream coherent for the store manager who must act on it.

6.3 Limitations

- **Fixed global α.** TAIN uses a single α for all channels. A per-channel or learned $\alpha_i(t)$ could improve performance, particularly for the retail 2–3 day gap reversal.
- **Tracking evaluation only.** TAIN is validated as a running statistics tracker, not yet integrated into end-to-end neural network training. Integration as a drop-in BatchNorm replacement in architectures such as TabNet or FT-Transformer is left to future work.
- **Jitter decomposition requires ground truth.** The unnecessary jitter metric requires the true signal x_t, limiting its use to evaluation settings. An online proxy (e.g., based on rolling statistics) would be needed for real-time monitoring.
- **Retail 2–3 day gap reversal.** TAIN shows a -6.4% reversal at the 2–3 day gap stratum in the Rossmann dataset, attributable to weekly seasonality. Domain-specific or learned $\alpha(t)$ would address this.
- **Strict monotonicity holds only in the Sensor domain.** The OU prediction that improvement grows with Δt is satisfied at the aggregate (largest stratum is best in four of five domains) but with within-domain non-monotonicities at intermediate strata (Section 5.2.6). Finance offers little gap-size variability (the weekend gap dominates), and ICU-Urine shows noise in middle strata. The clean monotonic signature requires a domain with both substantial gap-size variation and a relatively stationary signal between gaps.
- **Entity-level heterogeneity in ICU-Temp.** The Spearman correlation between per-patient mean gap size and per-patient improvement is significantly negative ($\rho = -0.139$, $p < 0.001$). This is verified to be a patient-acuity confound (mean gap correlates with SAPS-I at $\rho = -0.281$ and SOFA at $\rho = -0.375$; Section 5.2.7) rather than a failure of the OU discretization, but it indicates that TAIN's gains are not uniformly distributed across the cohort. A patient-stratified deployment would need to characterize who benefits before generalizing.

6.4 Future Work

- **End-to-end integration**: replace standard BatchNorm with TAIN in production neural network architectures (TabNet, FT-Transformer, NODE) and measure downstream task performance.
- **Learned $\alpha(t)$**: address the retail 2–3 day gap reversal by learning a context-dependent base decay rate rather than using a fixed α.
- **Per-channel damping**: extend TAIN to per-channel time-aware normalization with a diagonal damping matrix, enabling channel-specific viscosity (a direction we term HyperNetwork Viscosity).
- **Stability-gated actions**: combine time-aware normalization with a Lyapunov-motivated output gate that suppresses actions during detected instability (a direction we term Lyapunov-Gated Action).

- **Online jitter decomposition**: develop a proxy for unnecessary jitter that does not require ground truth, enabling real-time action stability monitoring.

7. Conclusion

We have introduced **Time-Aware Inertial Normalization (TAIN)**, a time-aware update rule for the running statistics maintained by normalization layers, in which the fixed EMA coefficient α is replaced with $\alpha^{\Delta t}$, adapting the forgetting rate to the actual elapsed time between observations. While the formula $\alpha^{\Delta t}$ is well established in time series analysis [Wright, 1986, Zumbach & Müller, 2001], its application to the running-statistics update underlying normalization layers is, to our knowledge, novel. We ground TAIN in the Ornstein-Uhlenbeck process, providing a principled theoretical foundation.

To evaluate TAIN's impact on action stability, we employed a **jitter decomposition analysis** that separates total action jitter into signal-tracking (desirable) and unnecessary (undesirable) components, complemented by flip rate and directional agreement metrics drawn from signal processing and econometrics. This analytical lens reveals the RMSE–jitter trade-off that existing metrics (RMSE alone, or total jitter alone) cannot capture.

Empirical validation across five real-world domains (5,409 entities, 659,325 observations) demonstrates that TAIN achieves statistically significant improvements over standard EMA ($p < 0.001$ in four of five domains). The largest gap stratum yields the largest improvement in four of five domains, with the Sensor domain exhibiting a clean monotonic increase from +7.6% to +67.6% consistent with the Ornstein-Uhlenbeck discretization. The L^1 jitter decomposition reveals that TAIN's increase in total jitter over Standard EMA is overwhelmingly signal-tracking rather than noise (98%/2% signal/unnecessary in Retail, $\geq 80\%$ signal across all five domains).

The empirical case for TAIN rests on three pillars:

1. **Statistical significance:** $p < 0.001$ in four of five domains with 3,684/5,409 entities (68%) showing per-entity improvement.
2. **Gap-proportional recovery:** the largest gap stratum yields the largest improvement in four of five domains; the Sensor domain shows a clean monotonic +7.6% $\rightarrow$ +67.6% progression directly confirming the OU discretization. Within-domain non-monotonicities at intermediate strata (Retail 2–3 day reversal, Finance weekend dominance, ICU-Urine middle-stratum noise) and entity-level heterogeneity (ICU-Temp $\rho = -0.139$, verified to be a patient-acuity confound via SAPS-I/SOFA correlations; Section 5.2.7) expose the limits of a single global α.
3. **Efficient signal tracking:** the L^1 decomposition shows that $\geq 80\%$ of TAIN's total-jitter increment over Standard EMA is signal-tracking across all five domains (98% in Retail). TAIN's unnecessary-share inflation over EMA is the smallest among RMSE-improving methods; Kalman and DEMA raise the unnecessary share by 3–6 pp in every domain and double the flip rate.

More broadly, TAIN serves as the normalization substrate for operational digital twins. In such architectures, a decision layer (whether an LLM, a rule engine, or a human operator) consumes the normalized signal to produce daily actions. If the normalization layer carries stale momentum after an observation gap, the digital twin loses fidelity to the physical store it mirrors, and every downstream action inherits that staleness. By resetting running statistics in proportion to elapsed time, TAIN keeps the digital twin synchronized with reality across irregular observation schedules.

In retail operations, every unnecessary action change carries a real cost: cancelled purchase orders, shelf reorganization labor, staff redeployment friction. These costs are not captured by RMSE or log-likelihood. By suppressing unnecessary jitter, TAIN-based normalization addresses an operational dimension of error — action stability — that prediction-loss metrics alone do not. Quantifying the dollar cost of this stability for a specific deployment remains future work.

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Table 3: Tracking performance across 7 methods and 5 domains ($\alpha = 0.95$). Bold indicates best within each J/R class. Methods are grouped by jitter profile.

Method	RMSE ↓	Jitter ↓	J/R ↓	Post-Gap ↓
Retail (n=50 stores)				
Std EMA	1878.0	98.9	0.053	1374.3
Linear EMA	1906.2	90.9	0.047	1437.0
TAIN	**1861.4**	120.3	0.067	**1237.8**
DEMA	**1786.2**	191.9	0.107	1269.3
Interp+EMA	1868.2	108.6	0.059	**1266.7**
Kalman Filter	**1553.1**	410.8	0.268	**960.9**
Holt ES	2521.4	381.1	0.148	1838.6
Sensor (n=12 stations)				
Std EMA	52.57	2.77	0.053	42.32
Linear EMA	52.76	2.76	0.052	44.70
TAIN	**52.24**	3.05	0.058	**37.12**
DEMA	**43.71**	4.88	0.112	**35.17**
Interp+EMA	52.37	2.88	0.055	39.34
Kalman Filter	**28.53**	7.29	0.256	**20.60**
Holt ES	74.04	15.54	0.210	54.33
Finance (n=5 tickers)				
Std EMA	9.18	0.48	0.053	5.88
Linear EMA	10.42	0.48	0.046	6.75
TAIN	**7.58**	0.69	0.091	**4.74**
DEMA	**6.32**	0.76	0.120	**3.85**
Interp+EMA	8.28	0.57	0.069	5.24
Kalman Filter	**3.66**	1.13	0.307	**2.23**
Holt ES	9.21	2.03	0.219	5.72
ICU-Temp (n=1,787 patients, PhysioNet 2012)				
Std EMA	0.80	0.04	0.054	0.69
Interp+EMA	**0.73**	0.06	0.087	**0.57**
TAIN	0.75	0.08	0.118	**0.51**
DEMA	**0.62**	0.07	0.121	0.58
Linear EMA	0.68	0.11	0.265	0.75
Holt ES	1.36	0.31	0.228	1.30
Kalman Filter	**0.37**	0.18	0.524	**0.30**
ICU-Urine (n=3,555 patients, PhysioNet 2012)				
Std EMA	183.58	9.81	0.053	136.31
TAIN	**176.89**	13.98	0.084	**121.83**
Interp+EMA	177.54	11.90	0.069	126.98
DEMA	**138.65**	16.66	0.117	**96.35**
Linear EMA	162.09	21.79	0.151	134.17
Holt ES	611.64	137.45	0.210	490.64
Kalman Filter	**120.64**	26.82	0.249	**67.29**

Table 4: Statistical significance of TAIN vs. Standard EMA.

Domain	N	RMSE Impr. (%)	95% CI	Wilcoxon p	Win Rate
Retail	50	+1.05	$[0.77, 1.33]$	$< 0.001^{***}$	40/50
Sensor	12	+0.62	$[0.51, 0.74]$	0.0002^{***}	12/12
Finance	5	+17.32	$[16.98, 17.67]$	0.031^{*}	5/5
ICU-Temp	1,787	+3.04	$[1.95, 4.13]$	$< 0.001^{***}$	1,088/1,787
ICU-Urine	3,555	+3.78	$[3.51, 4.06]$	$< 0.001^{***}$	2,539/3,555
Overall	5,409	+3.59	$[3.12, 4.07]$	—	3,684/5,409

Table 5: Stratified post-gap recovery: TAIN improvement over Standard EMA by gap size. Strata in italics have $n < 50$ gaps and are reported for completeness but have wide confidence; we draw conclusions from the well-populated strata.

Domain	Gap Stratum	n (gaps)	Post-Gap MAE Impr. (%)
Retail	1–2 days	3,575	+2.8
	2–3 days	186	−6.4
	7+ days	*5*	+41.8
Sensor	1–3 hours	2,315	+7.6
	3–6 hours	309	+18.0
	6–12 hours	88	+29.1
	12–24 hours	83	+43.6
	24+ hours	*42*	+67.6
Finance	2 days (Sat)	110	+29.0
	3 days (weekend)	2,260	+18.2
	4+ days (holiday)	350	+23.7
ICU-Temp	1.5–3 hours	4,739	+23.0
	3–6 hours	8,479	+34.4
	6–12 hours	749	+34.0
	12–24 hours	*16*	+39.3
	24+ hours	*1*	+36.8
ICU-Urine	1.5–3 hours	20,800	+13.4
	3–6 hours	3,951	+18.5
	6–12 hours	438	+16.0
	12–24 hours	*40*	+10.5
	24+ hours	*2*	+43.9

Table 6: Spearman rank correlation between per-entity mean gap size and per-entity RMSE improvement (TAIN vs. Standard EMA).

Domain	N entities	Spearman ρ	p-value
Retail	50	+0.422	0.007^{**}
Sensor	12	+0.126	0.697 (n.s.)
Finance	5	—	— (insufficient N)
ICU-Temp	1,787	−0.139	$< 0.001^{***}$
ICU-Urine	3,555	+0.248	$< 0.001^{***}$

Table 7: TAIN RMSE improvement (%) over Standard EMA across α values.

α	Retail (%)	Sensor (%)	Finance (%)
0.80	+4.05	+1.12	+18.27
0.85	+3.06	+1.02	+18.35
0.90	+2.03	+0.89	+17.97
0.95	+1.05	+0.62	+17.32
0.99	+0.47	+0.06	+18.73

Table 8: L^1 jitter decomposition across methods. Total, Signal, and Unnec. are L^1 means of absolute changes (additive). Unnec.% $= J^{L^1}_{\text{unnec}}/J^{L^1}_{\text{total}}$ (a strict fraction). RMS column reproduces Table 3 for reference.

	Method	Tot. L^1	Sig. L^1	Unnec. L^1	Unnec.%	Flip%	DirAgr%
Retail	Std EMA	75.88	74.52	1.36	**1.7%**	33.7%	66.5%
	TAIN	84.82	83.25	1.57	1.8%	33.5%	66.9%
	Linear EMA	67.30	65.98	1.32	2.0%	33.9%	66.2%
	Interp+EMA	80.32	78.88	1.44	1.7%	33.6%	66.7%
	Kalman Filter	302.03	287.41	14.63	4.7%	38.8%	72.2%
	DEMA	146.72	142.02	4.70	3.1%	34.4%	67.3%
	Holt ES	294.10	252.97	41.13	12.7%	10.6%	50.6%
Sensor	Std EMA	1.90	1.59	0.31	**16.4%**	9.7%	60.0%
	TAIN	1.92	1.61	0.32	16.4%	9.8%	60.1%
	Linear EMA	1.89	1.58	0.31	16.5%	9.7%	60.0%
	Interp+EMA	1.91	1.60	0.31	16.4%	9.7%	60.1%
	Kalman Filter	4.39	3.49	0.89	20.3%	18.4%	71.5%
	DEMA	3.21	2.49	0.72	22.3%	11.6%	62.8%
	Holt ES	10.81	5.21	5.60	51.8%	7.2%	50.4%
Finance	Std EMA	0.31	0.28	0.02	**7.9%**	8.6%	59.4%
	TAIN	0.37	0.33	0.04	11.5%	10.7%	61.3%
	Linear EMA	0.27	0.25	0.03	9.2%	7.6%	58.5%
	Interp+EMA	0.34	0.31	0.03	9.2%	9.5%	60.3%
	Kalman Filter	0.68	0.60	0.09	12.7%	20.9%	70.8%
	DEMA	0.47	0.42	0.05	11.4%	11.5%	62.0%
	Holt ES	1.27	0.86	0.41	32.1%	6.7%	52.7%
ICU-T	Std EMA	0.037	0.030	0.007	20.0%	12.6%	66.7%
	TAIN	0.058	0.049	0.009	**18.1%**	15.1%	68.4%
	Linear EMA	0.052	0.033	0.019	30.4%	17.0%	74.1%
	Interp+EMA	0.050	0.041	0.008	19.0%	14.4%	68.0%
	Kalman Filter	0.126	0.111	0.015	14.9%	23.9%	80.7%
	DEMA	0.062	0.050	0.012	20.4%	14.6%	69.0%
	Holt ES	0.272	0.150	0.122	39.0%	6.7%	55.7%
ICU-U	Std EMA	8.17	6.36	1.82	**14.8%**	18.3%	64.0%
	TAIN	9.98	7.82	2.16	15.0%	19.1%	64.7%
	Linear EMA	8.86	6.92	1.94	16.5%	19.1%	64.5%
	Interp+EMA	9.18	7.22	1.96	14.6%	18.7%	64.3%
	Kalman Filter	18.33	14.06	4.27	18.6%	30.3%	73.5%
	DEMA	12.70	9.42	3.28	17.8%	21.9%	66.4%
	Holt ES	118.38	36.30	82.07	46.9%	7.4%	51.7%

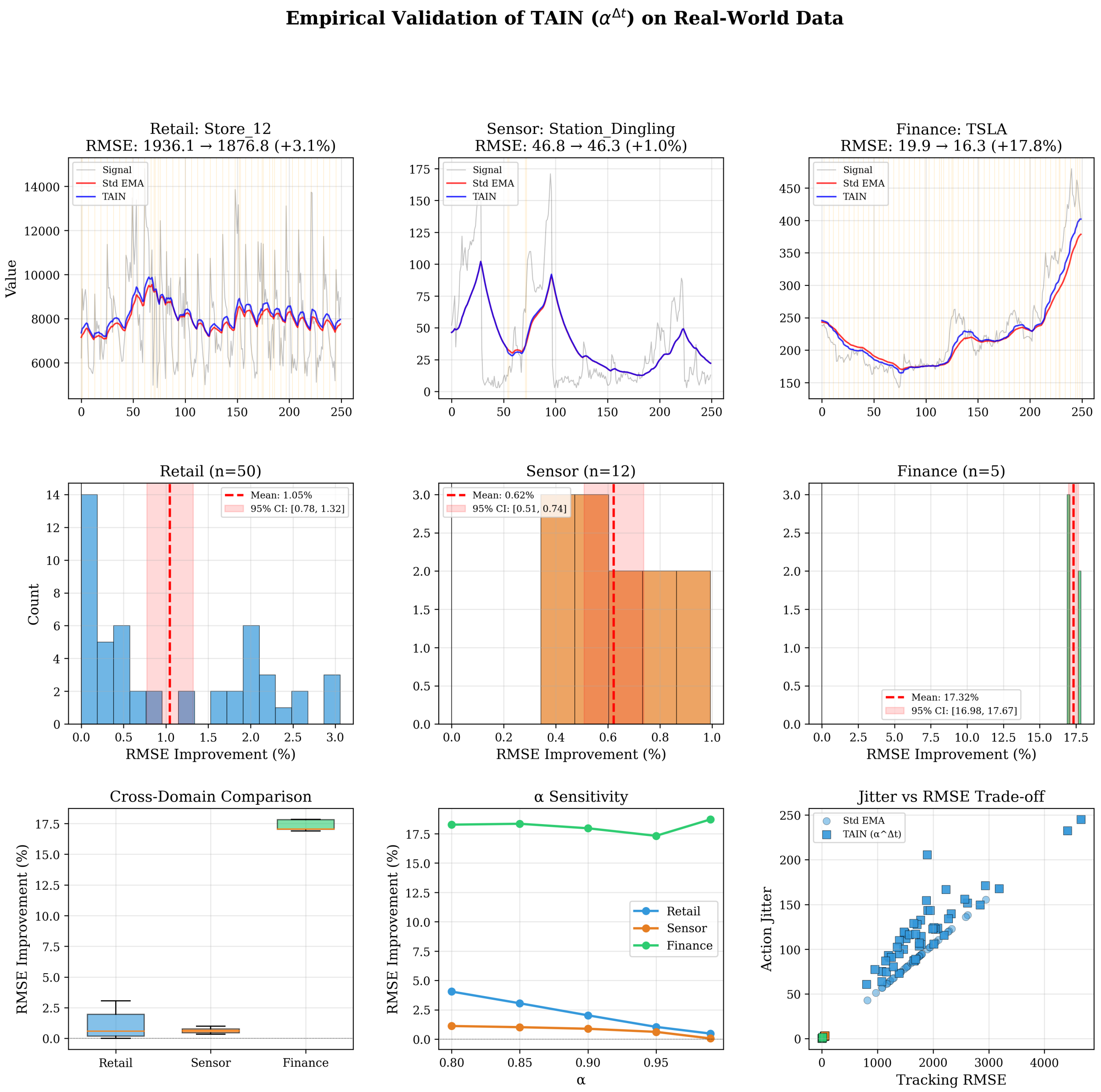

Figure 1: Comprehensive 7-method comparison across five domains. Top row: tracking examples with gap annotations. Middle row: RMSE distributions. Bottom row: box plots of J/R ratio, α sensitivity, and RMSE vs. Jitter trade-off.

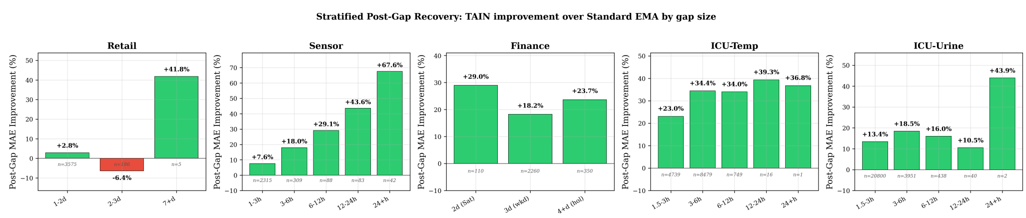

Figure 2: Stratified post-gap recovery: TAIN improvement (%) over Standard EMA as a function of gap size. The sensor domain exhibits a near-perfect monotonic relationship (7.6% → 67.6%), directly confirming the OU discretization prediction.

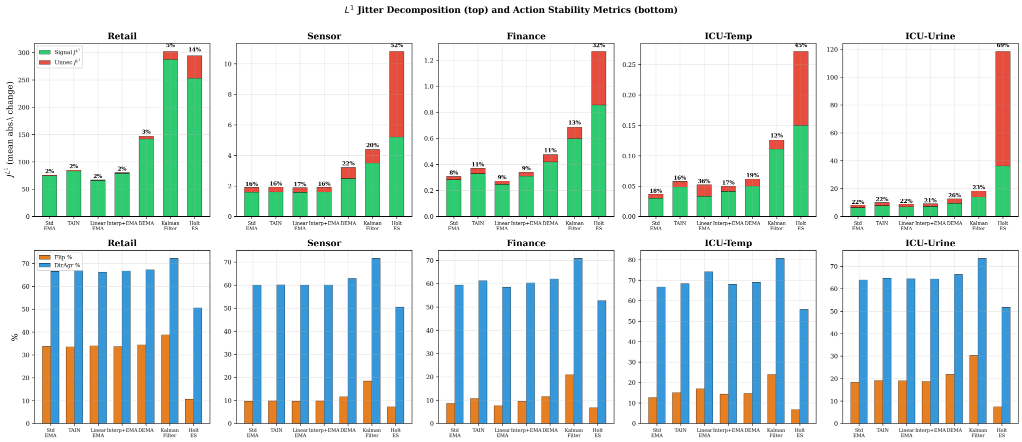

Figure 3: L^1 jitter decomposition. Left: stacked bar chart showing signal jitter (desirable) vs. unnecessary jitter (undesirable) for each method, with the unnecessary share annotated. Right: flip rate and directional agreement. Among RMSE-improving methods, TAIN has the smallest unnecessary-share inflation over Standard EMA.

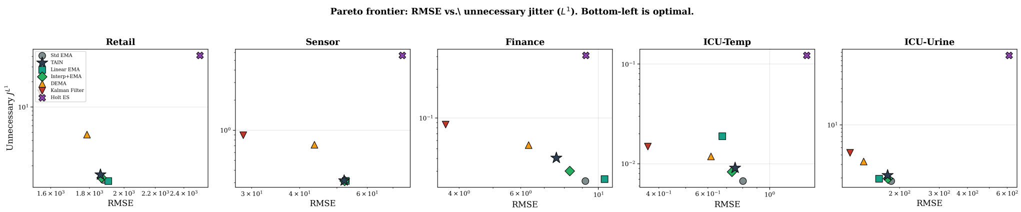

Figure 4: Pareto frontier: RMSE vs. unnecessary jitter. The bottom-left corner is optimal. TAIN occupies the Pareto-optimal region in the low-jitter regime; the Kalman Filter dominates the low-RMSE regime at the cost of substantially higher action instability.

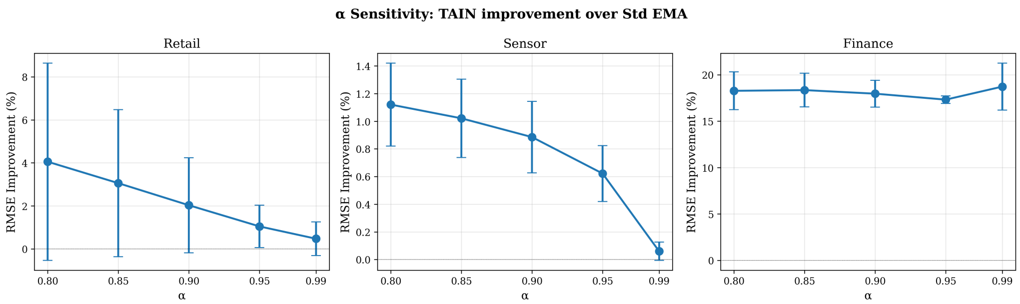

Figure 5: Sensitivity of TAIN's RMSE improvement to the base α parameter. Improvement is positive across all tested values ($\alpha \in [0.80, 0.99]$) in all five domains, with lower α producing larger improvements in Retail and Sensor.